

Trainings ▾

Preparatory Steps

With EGR

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)
- Remove the alternator drive belt. Refer to Procedure 013-005 in Section 13. (</qs3/pubsys2/xml/en/procedures/35/35-013-005-tr.html>)
- Remove the alternator. Refer to Procedure 013-001 in Section 13. (</qs3/pubsys2/xml/en/procedures/35/35-013-001-tr.html>).
- Remove the EGR cooler coolant supply upper tube. Refer to Procedure 011-031 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html>)
- Remove the water manifold return tube. Refer to Procedure 008-061 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-061-tr.html>)
- Remove the thermostat housing and bypass hose. Refer to Procedure 008-013 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-013-tr.html>)

Without EGR

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

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⚠ WARNING ⚠

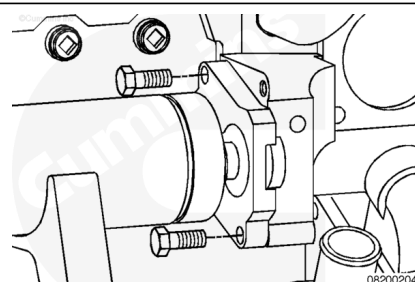
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)
- Remove the alternator drive belt. Refer to Procedure 013-005 in Section 13. (</qs3/pubsys2/xml/en/procedures/35/35-013-005-tr.html>)
- Remove the alternator drive belt and the alternator. Refer to Procedure 013-001 in Section 13. (</qs3/pubsys2/xml/en/procedures/35/35-013-001-tr.html>)
- Remove the thermostat housing and bypass hose. Refer to Procedure 008-015 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-015-tr.html>)

Remove

With EGR

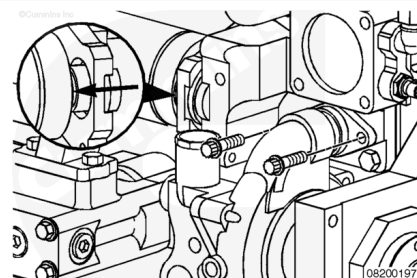
Remove the two oil cooler water connection retainer mounting flange capscrews.



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Push the water transfer tube connections into the lubricating oil cooler as far as possible.

Remove the two water transfer tube mounting capscrews.

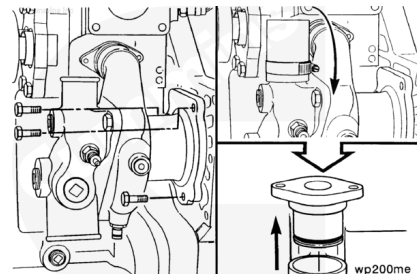


Remove the water pump.

Refer to Procedure 008-062 in Section 8.

(</qs3/pubsys2/xml/en/procedures/35/35-008-062-tr.html>)

Remove the water transfer tube from the water pump.

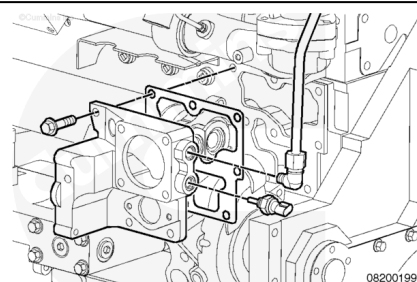


Remove the air compressor coolant return line. Refer to Procedure 012-109 in Section 12.

(</qs3/pubsys2/xml/en/procedures/35/35-012-109-tr.html>)

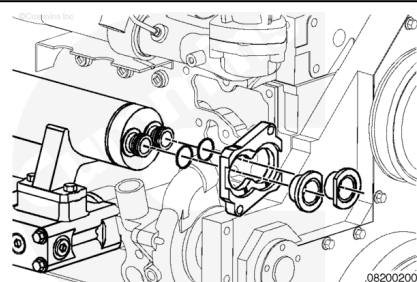
Remove the coolant temperature sensor. Refer to Procedure 019-019 in Section 19 (</qs3/pubsys2/xml/en/procedures/88/88-019-019.html>) in the Troubleshooting and Repair Manual, CM870 Electronic Control System, ISM Engines, Bulletin 4021381.

Remove the remaining capscrews and the thermostat housing support.



Remove the lubricating oil cooler water connection tubes from the cooler.

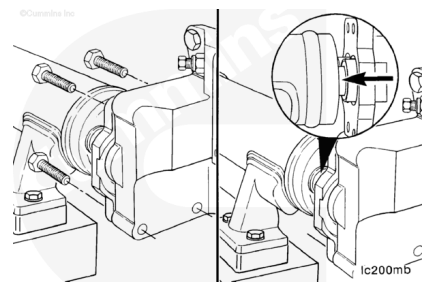
Remove and discard all four o-rings.



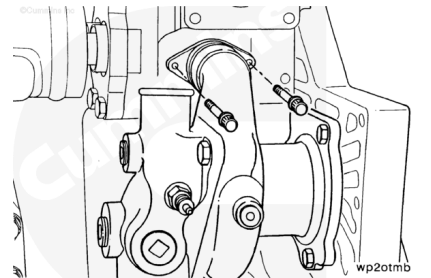
Without EGR

Remove the four lubricating oil cooler thermostat support flange capscrews.

Push the water transfer tube connections into the lubricating oil cooler as far as possible.

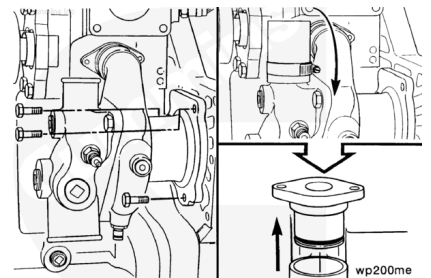


Remove the two water transfer tube mounting capscrews.



Remove the three water pump mounting capscrews.

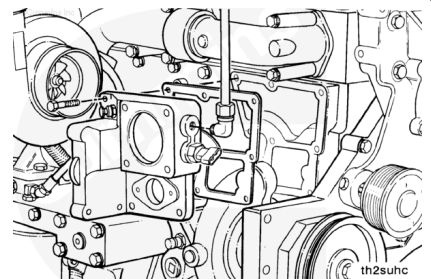
Twist the water pump outward, and remove the water transfer tube from the water pump body.



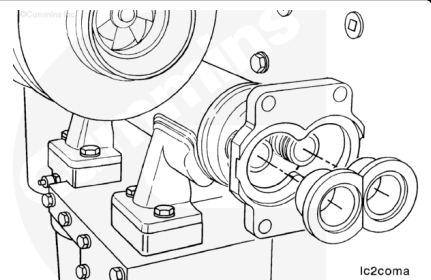
Remove the air compressor coolant return line.

Remove the coolant temperature sensor.

Remove the remaining capscrews and the thermostat housing support.



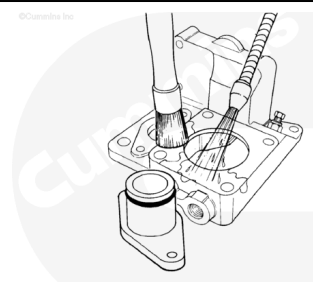
Remove the lubricating oil cooler water connection tubes from the cooler.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



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⚠ WARNING ⚠

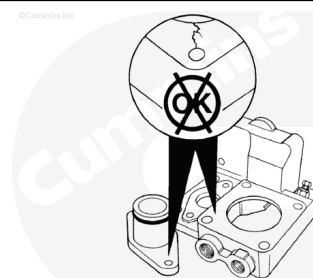
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean the lubricating oil cooler water transfer tubes, water pump water transfer tubes, and thermostat housing support.

Dry with compressed air.

Inspect the water transfer tubes and thermostat housing support for cracks or other damage.

If any parts are damaged, they **must** be replaced.



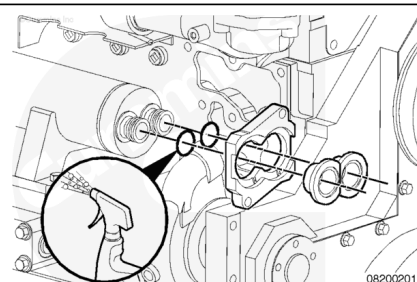
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Install

With EGR

⚠ CAUTION ⚠

Do not use petroleum products on the coolant plumbing o-rings. The o-rings will swell and cause the connection to leak.



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To avoid pinching or rolling the o-ring during assembly, do **not** lubricate the o-ring. Lubricate the o-ring **connection mating surfaces** with clean engine coolant, soapy water, or vegetable

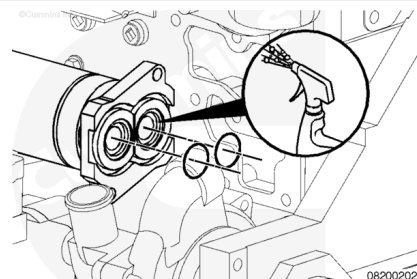
oil.

Install new o-rings on the lubricating oil cooler water connection tubes.

While holding the connection flange in place, install the water connections into the lubricating oil cooler. The straight edges of the connection **must** face each other.

⚠ CAUTION ⚠

Do not use petroleum products on the coolant plumbing o-rings. The o-rings will swell and cause the connection to leak.



To avoid pinching or rolling the o-ring during assembly, do **not** lubricate the o-ring. Lubricate the o-ring **connection mating surfaces** with clean engine coolant, soapy water, or vegetable oil.

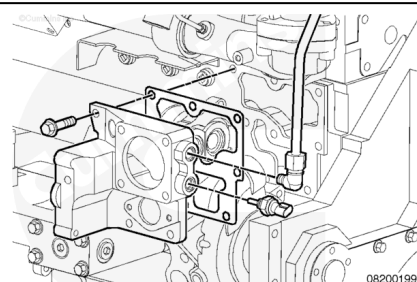
Install two new o-rings on the end of the lubricating oil cooler water tube connections.

Use a new gasket, and install the thermostat housing support and capscrews.

Torque Value: 54 n·m [40 ft-lb]

Install the coolant temperature sensor. Refer to Procedure 019-019 in Section 19 (</qs3/pubsys2/xml/en/procedures/88/88-019-019.html>) in the Troubleshooting and Repair Manual, CM870 Electronic Control System, ISM Engines, Bulletin 4021381.

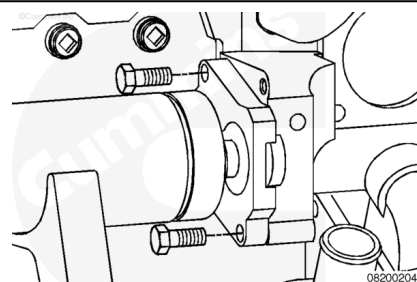
Install the air compressor coolant outline line. Refer to Procedure 012-109 in Section 12. (</qs3/pubsys2/xml/en/procedures/35/35-012-109-tr.html>)



Install the two oil cooler water connection retainer mounting flange capscrews.

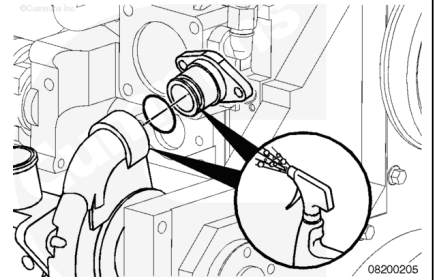
Torque Value: 47 n·m [35 ft-lb]

Make sure the o-rings on the end of the connection tubes are located correctly.



⚠ CAUTION ⚠

Do not use petroleum products on the coolant plumbing o-rings. The o-rings will swell and cause the connection to leak.

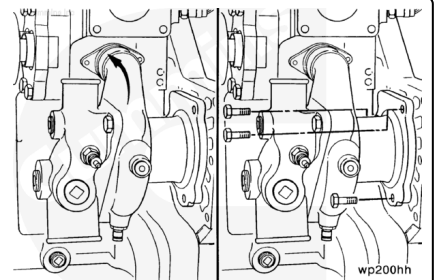


To avoid pinching or rolling the o-ring during assembly, do **not** lubricate the o-ring. Lubricate the o-ring **connection mating surfaces** with clean engine coolant, soapy water, or vegetable oil.

Install a new o-ring on the water pump water transfer tube.

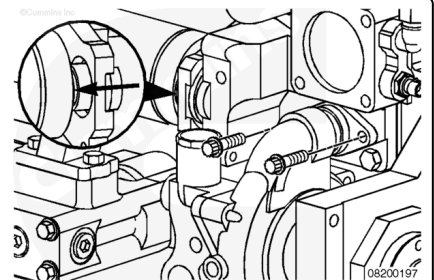
Install the water transfer tube and o-ring assembly in the water pump body.

Install the water pump. Refer to Procedure 008-062 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-062-tr.html>)



Use a new gasket, and install the water pump water transfer tube mounting capscrews.

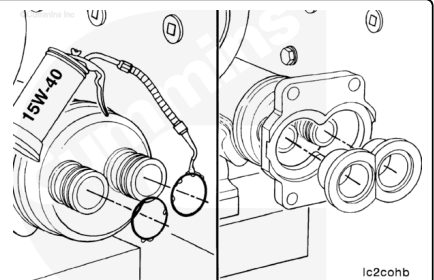
Torque Value: 24 n•m [212 in-lb]



Without EGR

⚠ CAUTION ⚠

Do not use petroleum products on the coolant plumbing o-rings. The o-rings will swell and cause the connection to leak.



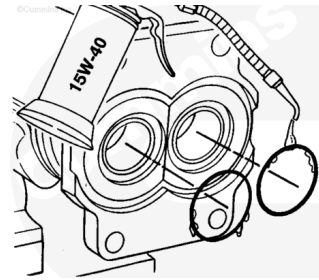
To avoid pinching or rolling the o-ring during assembly, do **not** lubricate the o-ring. Lubricate the o-ring **connection mating surfaces** with clean engine coolant, soapy water, or vegetable oil.

Install new o-rings on the lubricating oil cooler water connection tubes.

While holding the connection flange in place, install the water connections into the lubricating oil cooler. The straight edges of the connection **must** face each other.

⚠ CAUTION ⚠

Do not use petroleum products on the coolant plumbing o-rings. The o-rings will swell and cause the connection to leak.



To avoid pinching or rolling the o-ring during assembly, do **not** lubricate the o-ring. Lubricate the o-ring **connection mating surfaces** with clean engine coolant, soapy water, or vegetable oil.

Install two new o-rings on the end of the lubricating oil cooler water tube connections.

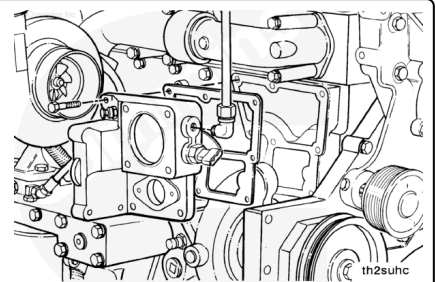
Use a new gasket, and install the thermostat housing support and capscrews.

Tighten the capscrews.

Torque Value: 54 n•m [40 ft-lb]

Install the coolant temperature sensor.

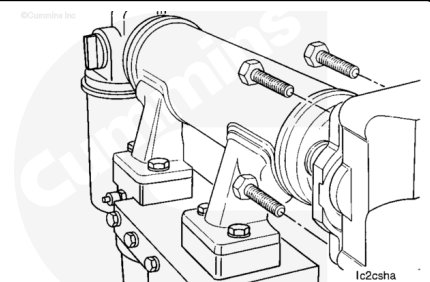
Install the air compressor coolant supply line.



Install the four mounting flange capscrews. Alternately and evenly, tighten the four capscrews.

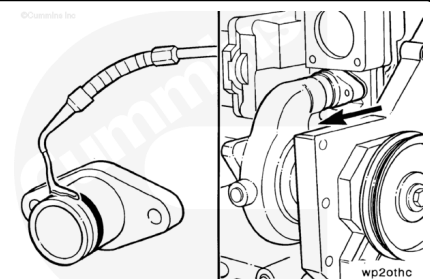
Torque Value: 47 n•m [35 ft-lb]

Be sure the o-rings on the end of the connection tubes are located correctly.



⚠ CAUTION ⚠

Do not use petroleum products on the coolant plumbing o-rings. The o-rings will swell and cause the connection to leak.



To avoid pinching or rolling the o-ring during assembly, do **not** lubricate the o-ring. Lubricate the o-ring **connection mating surfaces** with clean engine coolant, soapy water, or vegetable

oil.

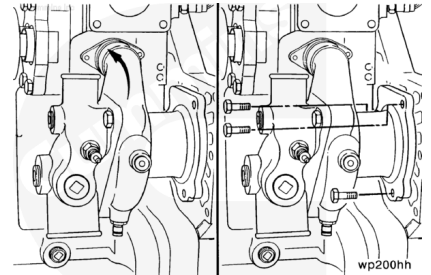
Install a new o-ring on the water pump water transfer tube.

Install the water transfer tube and o-ring assembly in the water pump body.

Rotate the water pump inward and install the three water pump mounting capscrews.

Tighten the capscrews.

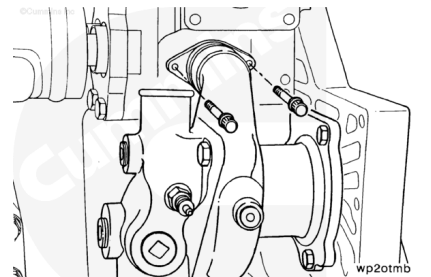
Torque Value: 47 n•m [35 ft-lb]



Use a new gasket and install the water pump water transfer tube mounting capscrews.

Tighten the capscrews.

Torque Value: 24 n•m [212 in-lb]



Finishing Steps

With EGR

- Install the thermostat and thermostat housing. Refer to Procedure 008-013 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-013-tr.html)
- Install the water manifold return tube. Refer to Procedure 008-061 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-061-tr.html)
- Install the EGR cooler coolant supply upper tube. Refer to Procedure 011-031 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html)
- Install the alternator. Refer to Procedure 013-001 in Section 13.
(/qs3/pubsys2/xml/en/procedures/35/35-013-001-tr.html)



- Install the alternator drive belt. Refer to Procedure 013-005 in Section 13. (</qs3/pubsys2/xml/en/procedures/35/35-013-005-tr.html>)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)
- Operate the engine until it reaches a temperature of 82°C [176°F] and check for coolant leaks.

Without EGR

- Install the thermostat and thermostat housing. Refer to Procedure 008-013 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-013-tr.html>)
- Install the alternator. Refer to Procedure 013-001 in Section 13. (</qs3/pubsys2/xml/en/procedures/35/35-013-001-tr.html>)
- Install the alternator drive belt. Refer to Procedure 013-005 in Section 13. (</qs3/pubsys2/xml/en/procedures/35/35-013-005-tr.html>)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)
- Operate the engine until it reaches a temperature of 82°C [176°F] and check for coolant leaks.

